

A Degree-Three Counterexample to Upward Validity for Steiner Cut Dominants

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Abstract

We give an explicit counterexample to the proposed upward-validity property for Steiner cut dominants. The example is a ten-vertex graph with terminal set $T = \{0, 1, \dots, 8\}$ and a single nonterminal vertex 9. The graph has no cut node and the only nonterminal has degree three. Nevertheless, a positive integer inequality defines a facet of $\text{CUT}^+(G, T)$ but is violated by the global cut $\delta(\{9\})$ after 9 is also made terminal. Thus upward validity fails even for an irreducible facet-inducing Steiner graph with a degree-three nonterminal.

1 Definitions and the facet test

Let $G = (V, E)$ be an undirected graph and let $T \subseteq V$ be a set of terminals. A cut $\delta(S)$ is a T -Steiner cut if

$$S \cap T \neq \emptyset \quad \text{and} \quad T \setminus S \neq \emptyset.$$

The T -Steiner cut dominant is

$$\text{CUT}^+(G, T) = \text{conv}\{\chi^{\delta(S)} : \delta(S) \text{ is a } T\text{-Steiner cut}\} + \mathbb{R}_{\geq 0}^E.$$

For a nonnegative edge-weight vector $c \in \mathbb{R}_{\geq 0}^E$, write

$$\gamma_c(G, T) = \min\{c(\delta(S)) : \delta(S) \text{ is a } T\text{-Steiner cut}\}.$$

The T -Steiner cuts of weight $\gamma_c(G, T)$ are called roots of c .

We use the standard facet test from Conforti and Kaibel [1]: if $cx \geq \gamma_c(G, T) > 0$ is valid, all edges have positive c -weight, and there are $|E|$ linearly independent roots, then the inequality defines a facet of $\text{CUT}^+(G, T)$. We also use their terminology that a facet-inducing Steiner graph (G, T) is *irreducible* if G has no cut node and no nonterminal vertex has degree two.

2 The counterexample

Let

$$V = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}, \quad T = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}.$$

Thus 9 is the only nonterminal. Let G have the following edge set and weights. The edge order in Table 1 is used throughout the proof.

Consider the inequality

$$\begin{aligned} &2x_{02} + 3x_{04} + x_{12} + 3x_{17} + 3x_{26} + 2x_{34} + 3x_{35} + x_{48} \\ &+ 2x_{57} + x_{58} + 3x_{68} + x_{78} + x_{09} + x_{39} + 2x_{19} \geq 6. \end{aligned} \tag{1}$$

e	02	04	12	17	26	34	35	48	57	58	68	78	09	39	19
c_e	2	3	1	3	3	2	3	1	2	1	3	1	1	1	2

Table 1: The edge set and weights of the counterexample.

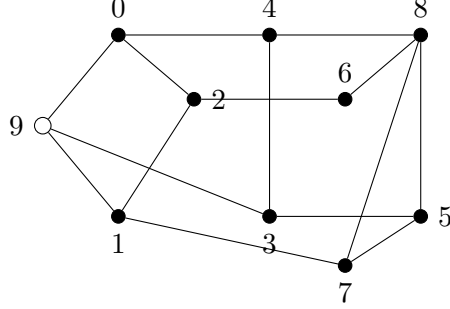


Figure 1: The graph G . Black vertices are terminals; vertex 9 is the only nonterminal. Weights are listed in Table 1.

Theorem 1. *The inequality (1) is facet-defining for $\text{CUT}^+(G, T)$, the Steiner graph (G, T) is irreducible, and (1) is not valid for the global cut dominant $\text{CUT}^+(G, V)$.*

3 Validity for $\text{CUT}^+(G, T)$

Let $H = G[T]$ be the subgraph induced by the terminals. For $A \subseteq T$, put $d_H(A) = c(\delta_H(A))$. The only edges incident with the nonterminal 9 are 09, 39, 19, with weights 1, 1, 2. Therefore, if $S \subset V$ and $A = S \cap T$, then the contribution of the three edges incident with 9 is either

$$r(A) = \mathbf{1}_{0 \in A} + 2\mathbf{1}_{1 \in A} + \mathbf{1}_{3 \in A}$$

if $9 \notin S$, or $4 - r(A)$ if $9 \in S$. Hence every cut with terminal side A has weight at least

$$d_H(A) + h(A), \quad h(A) = \min\{r(A), 4 - r(A)\}.$$

The following table lists, up to replacing A by $T \setminus A$, all terminal cuts of H with $d_H(A) < 6$. The completeness of this short finite list is verified by the exact enumeration in Appendix B.

A	$d_H(A)$	$h(A)$	$d_H(A) + h(A)$
$\{0\}$	5	1	6
$\{1\}$	4	2	6
$\{3\}$	5	1	6
$\{0, 4\}$	5	1	6
$\{3, 5\}$	5	1	6
$\{1, 7\}$	4	2	6
$\{1, 3, 5, 7\}$	5	1	6

Table 2: All terminal cuts of $H = G[T]$ of weight less than 6, up to complement.

For every other nonempty proper $A \subset T$, the same enumeration gives $d_H(A) \geq 6$, and hence $d_H(A) + h(A) \geq 6$. Consequently every T -Steiner cut has c -weight at least 6. Thus (1) is valid for

$\text{CUT}^+(G, T)$ and

$$\gamma_c(G, T) = 6.$$

4 A root basis

The following fifteen T -Steiner cuts all have c -weight 6:

$$\begin{aligned} &\{0\}, \{1\}, \{2\}, \{3\}, \{4\}, \{0, 4\}, \{5\}, \{3, 5\}, \\ &\{6\}, \{2, 6\}, \{7\}, \{1, 7\}, \{8\}, \\ &\{0, 2, 4, 6, 8\}, \quad \{0, 2, 3, 4, 5, 6, 8\}. \end{aligned}$$

With respect to the edge order in Table 1, the matrix whose rows are the incidence vectors of these cuts is displayed in Appendix A. Its determinant is

$$192 \neq 0.$$

Thus the fifteen roots are linearly independent. Since $|E| = 15$ and all edge weights are positive, the facet test recalled above implies that (1) defines a facet of $\text{CUT}^+(G, T)$.

5 Irreducibility

The graph has no cut node. Indeed, it is obtained from the cycle

$$0 - 2 - 1 - 9 - 0$$

by adding the ears

$$0 - 4 - 3 - 9, \quad 2 - 6 - 8 - 4, \quad 3 - 5 - 7 - 1,$$

and then the one-edge ears 58 and 78. Starting from a cycle, adding an ear whose endpoints lie in the current graph and whose internal vertices are new preserves two-vertex-connectivity. Hence G is two-vertex-connected, and in particular has no cut node.

The only nonterminal is 9, and

$$\deg_G(9) = 3.$$

Thus no nonterminal has degree two. Therefore (G, T) is irreducible in the sense of Conforti and Kaibel.

6 Failure of upward validity

If vertex 9 is also made terminal, then the singleton cut $\delta(\{9\})$ is a valid global cut. Its c -weight is

$$c(\delta(\{9\})) = c_{09} + c_{39} + c_{19} = 1 + 1 + 2 = 4.$$

This is strictly smaller than the right-hand side 6 in (1). Therefore (1) is not valid for $\text{CUT}^+(G, V)$. This proves Theorem 1 and gives a negative answer to upward validity, even with a degree-three nonterminal.

A The root matrix

With the edge order

02, 04, 12, 17, 26, 34, 35, 48, 57, 58, 68, 78, 09, 39, 19,

the root matrix is

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

An exact Bareiss elimination, reproduced in Appendix B, gives $\det(M) = 192$.

B Exact verification script

The following Python script uses only integer arithmetic. It verifies validity, the list of low terminal cuts in Table 2, the root weights, and the determinant of the root matrix.

```
V = set(range(10))
T = set(range(9))

E = [
    (0,2), (0,4), (1,2), (1,7), (2,6),
    (3,4), (3,5), (4,8), (5,7), (5,8),
    (6,8), (7,8), (0,9), (3,9), (1,9)
]
c = [2,3,1,3,3,2,3,1,2,1,3,1,1,1,2]

roots = [
    {0}, {1}, {2}, {3}, {4}, {0,4}, {5}, {3,5},
    {6}, {2,6}, {7}, {1,7}, {8},
    {0,2,4,6,8},
    {0,2,3,4,5,6,8}
]

def cut_vector(S):
    return [int((u in S) ^ (v in S)) for u, v in E]

def weight(S):
    return sum(ci * xi for ci, xi in zip(c, cut_vector(S)))

def terminal_cut_weight(A):
    A = set(A)
    total = 0
    for (u,v), ci in zip(E, c):
        if u in T and v in T and ((u in A) ^ (v in A)):
            total += ci
    return total

def h(A):
    A = set(A)
    r = (1 if 0 in A else 0) + 2*(1 if 1 in A else 0) + (1 if 3 in A else 0)
    return min(r, 4-r)
```

```

def det_bareiss(A):
    A = [list(row) for row in A]
    n = len(A)
    if n == 0:
        return 1
    sign = 1
    previous = 1
    for k in range(n-1):
        pivot = k
        while pivot < n and A[pivot][k] == 0:
            pivot += 1
        if pivot == n:
            return 0
        if pivot != k:
            A[k], A[pivot] = A[pivot], A[k]
            sign = -sign
        pivot_value = A[k][k]
        for i in range(k+1, n):
            for j in range(k+1, n):
                A[i][j] = (A[i][j]*pivot_value - A[i][k]*A[k][j]) // previous
        previous = pivot_value
        for i in range(k+1, n):
            A[i][k] = 0
    return sign * A[n-1][n-1]

# Validity for all T-Steiner cuts.
values = []
for mask in range(1, 1 << 10):
    S = {i for i in range(10) if (mask >> i) & 1}
    if S == V:
        continue
    if S & T and T - S:
        values.append(weight(S))
assert min(values) == 6

# Low terminal cuts of  $H=G[T]$ , up to complement.
all_terminals = (1 << 9) - 1
low = []
for mask in range(1, all_terminals):
    comp = all_terminals ^ mask
    if mask > comp:
        continue
    A = {i for i in range(9) if (mask >> i) & 1}
    d = terminal_cut_weight(A)
    if d < 6:
        low.append((tuple(sorted(A)), d, h(A), d + h(A)))

expected_low = [
    ((0,), 5, 1, 6),
    ((1,), 4, 2, 6),
    ((3,), 5, 1, 6),
    ((0,4), 5, 1, 6),
    ((3,5), 5, 1, 6),
    ((1,7), 4, 2, 6),
    ((1,3,5,7), 5, 1, 6),
]
assert low == expected_low

# Root weights and linear independence.
assert all(weight(S) == 6 for S in roots)
M = [cut_vector(S) for S in roots]
assert det_bareiss(M) == 192

# Upward-validity failure.
assert weight({9}) == 4
print("All checks passed.")

```

References

- [1] Michele Conforti and Volker Kaibel. Steiner cut dominants. *Mathematics of Operations Research*, 50(1):764–781, 2025. arXiv:2209.14802.